Surge & Deaerator (DA) Tanks: Process, Flow & Optimization Overview

Purpose:

Surge tanks provide additional volume buffering for fluctuating loads, while DA tanks remove dissolved oxygen and other gases from feedwater to protect boilers from corrosion.
Often used together in boiler systems to maintain consistent water levels, temperature, and oxygen-free feedwater supply.

System Start-to-Finish Flow:

1. **Condensate Collection (Surge Tank)**
- Condensate from steam traps returns to the surge tank.
- Helps manage sudden surges in return water without overloading the DA tank.

2. **Transfer to Deaerator Tank**
- Pump transfers condensate from surge to DA tank as needed.
- DA tank is typically mounted higher and pressurized.

3. **Deaeration Process**
- Incoming water is sprayed into a heated steam environment.
- Dissolved gases are vented off via a scrubber stack.
- Treated water drops into DA storage section, ready for use.

4. **Feedwater Supply to Boiler**
- Feedwater pumps send oxygen-free water to the boiler.
- Pressure, temperature, and water level are tightly monitored.

Key Components & Definitions:

- **Surge Tank**: Manages fluctuating returns before DA.
- **Deaerator**: Removes O_2 and CO_2 via temperature and mechanical action.
- **Scrubber Stack**: Vents liberated gases.
- **Level Controls**: Maintain correct fill levels across both tanks.
- **Feedwater Pump**: Pushes treated water to boiler.

Example Flow:

> In a process facility, three returns from steam traps feed into a 500-gallon surge tank. Based on float level, water is pumped to a 20 psi pressurized DA tank, where steam preheats to 212°F, removing oxygen. A 3 HP feedwater pump maintains 60 psi supply to a 200 HP firetube boiler.

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